

SHEEL PATEL

+91 97125 38315 | 2303031240960@paruluniversity.ac.in | linkedin.com/in/sheel-patel-a3939b287 | github.com/Sheel34 |
Open to Relocation

ACHIEVEMENTS

- National Winner — Limitless Youth India Hackathon 2025** 2025
- Ranked **1st out of 10,000+ teams** nationwide in the Science & Technology category; directly selected and invited by the organising committee — no open application
 - Robotics system received **national-level recognition from the Prime Minister of India**; evaluated by a jury comprising the **Chairman of ISRO**, NITI Aayog members, and DRDO technologists
 - Project recognised as a breakthrough in real-time robotic control systems for defence and hazardous-environment applications

PROJECTS

- Terrain Intelligence System** | *Python, NumPy, OpenCV, FastAPI, React, Three.js* | [GitHub](#) Jan 2025 – Present
- Engineered a multi-layer **analytical terrain-risk pipeline** computing slope (gradient analysis), surface roughness (sliding-window standard deviation), curvature (Laplacian), and a shadow proxy (hillshading via surface-normal dot product), then combining these into a weighted hazard score per grid cell
 - Implemented a **landing zone ranking algorithm** that scores and classifies candidate regions by combined hazard, traversability, and safety threshold metrics, outputting structured JSON for downstream decision support
 - Built a **FastAPI REST backend** exposing `/analyze`, `/analyze-upload`, and `/samples` endpoints; ingests PNG, JPEG, and **GeoTIFF** (via Rasterio) files up to 50 MB, normalises spatial grids, and returns fully typed analytical payloads
 - Developed a **React + Three.js 3D mission-control dashboard** with configurable terrain overlay layers (hazard, slope, roughness), interactive landing zone selection, lander physics simulation, audio telemetry, and downloadable mission reports — translating raw analytical data into actionable visual insights
 - Validated against procedural analogues of Mars Jezero, Mars Gale Crater, and Lunar South Pole datasets; maintained module-level technical documentation throughout
- EOD Robotic Arm Control System (VENOM)** | *C++, Python, ESP32, OpenCV* | [GitHub](#) Nov 2024 – Feb 2025
- Engineered a 5-DOF teleoperated robotic arm for Explosive Ordnance Disposal that replicates human hand kinematics in real time, reducing simulated task completion time by **62%**
 - Designed a custom operator glove integrating 10+ flex sensors and an MPU-6050 IMU with an ESP32; achieved end-to-end wireless control latency of ~20 ms over the ESP-NOW protocol
 - Implemented multithreaded FreeRTOS firmware for deterministic task scheduling enabling parallel control of 12+ servos; integrated an inverse kinematics solver achieving sub-centimetre positioning accuracy
 - System attained a **74% improvement** in EOD simulation success rates; drew active interest from **3+ national defence research institutions**

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C, C++, Java

Data & Analytics: NumPy, Pandas, OpenCV, spatial data analysis, slope/curvature/roughness computation, GeoTIFF processing (Rasterio), statistical signal processing, SQL (foundational)

Dashboards & Visualisation: React (Vite PWA), Three.js (3D visualisation), FastAPI REST APIs, JSON telemetry pipelines, data-driven UI design

Automation & Deployment: REST API design, file ingestion pipelines, task automation, Git, Linux, Docker (familiar), Azure (familiar), Agile methodology

Embedded Systems & Robotics: ESP32, FreeRTOS, PWM, I2C, SPI, ESP-NOW, Inverse Kinematics, Servo Control, Sensor Interfacing, MPU-6050

Platforms: ServiceNow (CSA Certified), GitHub, Rasterio, Pillow

CS Fundamentals: Object-Oriented Programming, Operating Systems, Computer Networks, DBMS

CERTIFICATIONS

- ServiceNow Certified System Administrator (CSA)** | *ServiceNow* 2025
- Certified in platform administration, automated workflow design, and ITSM configuration; demonstrates ability to deploy and maintain enterprise software solutions at scale

EDUCATION

Parul University <i>B.Tech — Artificial Intelligence & Data Science</i>	Vadodara, Gujarat <i>July 2023 – May 2027</i>
SSV 2 School <i>GHSEB — Physics, Chemistry, Mathematics (PCM) — 80%</i>	Vadodara, Gujarat <i>June 2021 – April 2023</i>